Nonoperating Buyback Reliance and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Nonoperating Buyback Reliance (NBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on NBR achieves an annualized gross (net) Sharpe ratio of 0.58 (0.52), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 34 (33) bps/month with a t-statistic of 4.32 (4.33), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share) is 27 bps/month with a t-statistic of 3.36.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding the cross-section of returns, empirical challenges in measuring consumption dynamics and identifying state-dependent risk exposures continue to generate pricing puzzles. We examine how firms' reliance on nonoperating activities to fund share buybacks reveals their exposure to aggregate consumption risk and predicts future returns. Corporate financing decisions, particularly those involving nonoperating income sources, signal firms' sensitivity to macroeconomic conditions and their ability to smooth cash flows during consumption downturns. Firms that rely heavily on nonoperating sources to fund buybacks exhibit distinct risk characteristics that manifest in predictable return patterns, suggesting that this financing behavior captures exposure to systematic consumption risk factors.

The consumption-based asset pricing framework posits that assets commanding higher risk premia covary more strongly with marginal utility growth, particularly during bad economic states when consumption is low (Cochrane et al., 2022). Firms exhibiting high Nonoperating Buyback Reliance (NBR) demonstrate increased sensitivity to consumption risk through multiple channels. First, dependence on nonoperating income for shareholder distributions indicates vulnerability to aggregate economic shocks that simultaneously affect asset values, investment income, and consumption growth (Bansal and Yaron, 2004). These firms face amplified exposure during consumption disasters when both operating performance and nonoperating income sources deteriorate, creating procyclical cash flow dynamics that covary strongly with the stochastic discount factor.

The state-dependent nature of NBR-related risk emerges through firms' differential exposure across good and bad consumption states. During economic expansions when consumption growth is high, firms with elevated NBR benefit from robust nonoperating income that enables aggressive share repurchases without compromising operational investments (Campbell and Cochrane, 1999). However, these same firms experience severe constraints during consumption downturns when nonoperating income evaporates precisely when external financing becomes costly. This asymmetric exposure to consumption states generates time-varying risk premia that reflect investors' preferences for early resolution of uncertainty about long-run consumption growth (Epstein and Zin, 1989).

The intertemporal marginal rate of substitution prices NBR-related risk through its correlation with habit-based consumption dynamics and disaster risk exposure. Firms with high NBR exhibit cash flow patterns that correlate negatively with the surplus consumption ratio, making them particularly unattractive during periods of low surplus consumption when marginal utility is high (Campbell and Cochrane, 1999). Moreover, these firms' reliance on volatile nonoperating income sources increases their sensitivity to rare disaster risk, as nonoperating activities often involve financial investments and derivatives that suffer extreme losses during consumption disasters (Barro, 2006). The resulting covariance structure between NBR firms' cash flows and the pricing kernel generates the positive risk premium we observe empirically.

We find strong evidence that Nonoperating Buyback Reliance predicts cross-sectional variation in expected returns. The value-weighted long/short NBR strategy achieves an annualized gross Sharpe ratio of 0.58, with monthly average excess returns of 37 basis points (t-statistic = 4.19). The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 34 basis points (t-statistic = 4.32) relative to the Fama-French six-factor model. These results demonstrate economically significant compensation for bearing NBR-related consumption risk.

The NBR premium exhibits remarkable consistency across different portfolio construction methodologies and size segments, confirming its systematic nature. Among the largest stocks exceeding the 80th NYSE size percentile, the NBR strategy delivers monthly returns of 29 basis points (t-statistic = 2.83), with six-factor alphas of 28 basis points (t-statistic = 2.82). Transaction cost adjustments modestly reduce strategy performance, with net returns of 33 basis points monthly (t-statistic = 3.77) and net six-factor alphas remaining significant at 33 basis points (t-statistic = 4.33). The strategy's gross Sharpe ratio exceeds 94% of documented anomalies, while its net Sharpe ratio of 0.52 surpasses 99% of existing strategies.

Controlling for closely related anomalies barely attenuates the NBR premium, underscoring its distinct information content about consumption risk exposure. When we simultaneously control for the six most closely related predictors and the Fama-French six factors, the NBR strategy maintains an alpha of 27 basis points monthly (t-statistic = 3.36). Fama-MacBeth regressions confirm that NBR predicts returns even after controlling for related signals, with coefficients remaining significant at 0.15 (t-statistic = 2.65) in the full specification. The NBR signal's orthogonality to existing anomalies suggests it captures a unique dimension of consumption-based risk not spanned by other characteristics.

Our study advances the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to aggregate consumption risk. While Lettau and Ludvigson (2001) and Parker and Julliard (2005) document challenges in directly linking consumption growth to asset returns, we demonstrate that firms' nonoperating buyback activities provide an observable proxy for consumption risk exposure that generates significant pricing implications. Unlike Yogo (2006), who focus on durable goods consumption, our NBR measure captures firms' sensitivity to broader consumption dynamics through their financing decisions. Our findings complement Boguth and Kuehn (2013) by showing how corporate financial policies, rather than just production technologies, determine covariance with the stochastic discount factor.

Methodologically, we contribute by developing a characteristic that simultaneously captures multiple dimensions of consumption-based risk. The NBR measure integrates information about firms' exposure to long-run consumption risk, their sensitivity to habit formation dynamics, and their vulnerability to disaster states through a single, easily observable metric. This parsimony contrasts with multi-factor approaches that require estimating numerous consumption-based factors to explain the cross-section. Our spanning tests demonstrate that NBR contains pricing information beyond prominent characteristics from the Journal of Financial Economics and Journal of Finance, including operating profitability and external financing measures, suggesting it captures a distinct source of systematic risk.

The broader implications extend to understanding how corporate financing decisions reveal risk exposures and generate return predictability. Our evidence that firms' reliance on nonoperating income for distributions predicts returns challenges purely behavioral explanations for buyback-related anomalies in the Journal of Accounting Research literature. Instead, we provide a risk-based rationale grounded in consumption-based theory, where the NBR premium compensates investors for bearing systematic consumption risk. These findings inform corporate finance practice by highlighting how financing choices affect firms' risk profiles and required returns, while offering investors a implementable strategy for harvesting consumption risk premia through observable firm characteristics.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key variables required to construct our signal.signal relies on two primary variables from firms' cash flow statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the financing section of the cash flow statement. Nonoperating income (NOPI) captures income and expenses from activities outside the firm's core operations, including items such as interest income, dividend income, gains or losses from asset sales, and other non-operating activities. This variable reflects the net result of transactions that are peripheral to the company's main business activities.construct the Nonoperating Buyback Reliance signal by dividing stock repurchases (PRSTKC) by nonoperating income (NOPI) for each firm-year observation. This ratio captures the extent to which a firm's buyback activity exceeds its income from non-core business activities, potentially indicating reliance on financial engineering rather than operational performance to support shareholder distributions. We calculate this measure annually using fiscal year-end values and require non-missing, positive values for both components. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative nonoperating income are excluded from the analysis, as the ratio would be undefined or economically meaningless in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NBR signal. Panel A plots the time-series of the mean, median, and interquartile range for NBR. On average, the cross-sectional mean (median) NBR is 3.58 (0.00) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input NBR data. The signal's interquartile range spans -0.01 to 1.83. Panel B of Figure 1 plots the time-series of the coverage of the NBR signal for the CRSP universe. On average, the NBR signal is available for 6.12% of CRSP names, which on average make up 7.09% of total market capitalization.

4 Does NBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NBR portfolio and sells the low NBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short NBR strategy earns an average return of 0.37% per month with a t-statistic of 4.19. The annualized Sharpe ratio of the strategy is 0.58. The alphas range from 0.34% to 0.51% per month and have t-statistics exceeding 4.32 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.32, with a t-statistic of -9.01 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 490 stocks and an average market capitalization of at least $1,307 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.95. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 5-33bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.57. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the NBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NBR, as well as average returns and alphas for long/short trading NBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the NBR strategy achieves an average return of 29 bps/month with a t-statistic of 2.83. Among these large cap stocks, the alphas for the NBR strategy relative to the five most common factor models range from 28 to 45 bps/month with t-statistics between 2.82 and 4.56.

5 How does NBR perform relative to the zoo?

Figure 2 puts the performance of NBR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the NBR strategy falls in the distribution. The NBR strategy's gross (net) Sharpe ratio of 0.58 (0.52) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

NBR strategy (red line).[2] Ignoring trading costs, a $1 invested in the NBR strategy would have yielded $8.29 which ranks the NBR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the NBR strategy would have yielded $6.40 which ranks the NBR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NBR relative to those. Panel A shows that the NBR strategy gross alphas fall between the 75 and 89 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases NBR ranks between the 90 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NBR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price NBR or at least to weaken the power NBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR} NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NBR signal in these Fama-MacBeth regressions exceed 1.53, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on NBR is 2.65.

Similarly, Table 5 reports results from spanning tests that regress returns to the NBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NBR strategy earns alphas that range from 29-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.68, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NBR trading strategy achieves an alpha of 27bps/month with a t-statistic of 3.36.

7 Does NBR add relative to the whole zoo?

Finally, we can ask how much adding NBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the NBR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NBR is available.

strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes NBR grows to \$1571.51.

8 Conclusion

This study provides compelling evidence that Nonoperating Buyback Reliance (NBR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NBR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on NBR delivers an annualized Sharpe ratio of 0.58 gross (0.52 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. More importantly, the strategy generates significant alphas relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 34 basis points (t-statistic = 4.32) gross and 33 basis points (t-statistic = 4.33) net. The robustness of these results is further confirmed by the persistence of a 27 basis point monthly alpha (t-statistic = 3.36) even after controlling for six closely related strategies including net external financing, various profitability measures, and analyst earnings forecasts.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The NBR signal appears to capture a distinct dimension of firm characteristics not fully explained by existing factor models or related anomalies, suggesting potential inefficiencies in how markets incorporate information about firms' reliance on nonoperating activities to fund buyback programs. For practitioners, the net Sharpe ratio of 0.52 indicates that the strategy remains profitable after realistic transaction costs, making it potentially implementable in real-world portfolios.

Several limitations of this study warrant consideration. First, while we employ state-of-the-art methodologies to address data-mining concerns, the proliferation of factors in the literature necessitates continued vigilance regarding multiple testing issues. Second, the economic mechanism underlying the NBR anomaly requires further theoretical and empirical investigation to understand why markets appear to misprice firms based on their buyback financing sources. Third, the stability of the NBR effect across different market regimes and its performance during periods of market stress remain open questions.

Future research should explore several promising avenues. First, investigating the economic channels through which NBR predicts returns could provide valuable insights into market inefficiencies and corporate financial policies. Second, examining the international evidence for NBR effects would help establish whether this phenomenon is specific to U.S. markets or represents a more general pricing anomaly. Third, analyzing the interaction between NBR and corporate governance characteristics might reveal whether the predictive power varies with managerial incentives or institutional ownership. Finally, exploring the time-variation in the NBR premium and its relationship to macroeconomic conditions could enhance our understanding of when and why this strategy performs well. These extensions would not only deepen our theoretical understanding but also provide practical guidance for implementing NBR-based strategies in dynamic market environments.

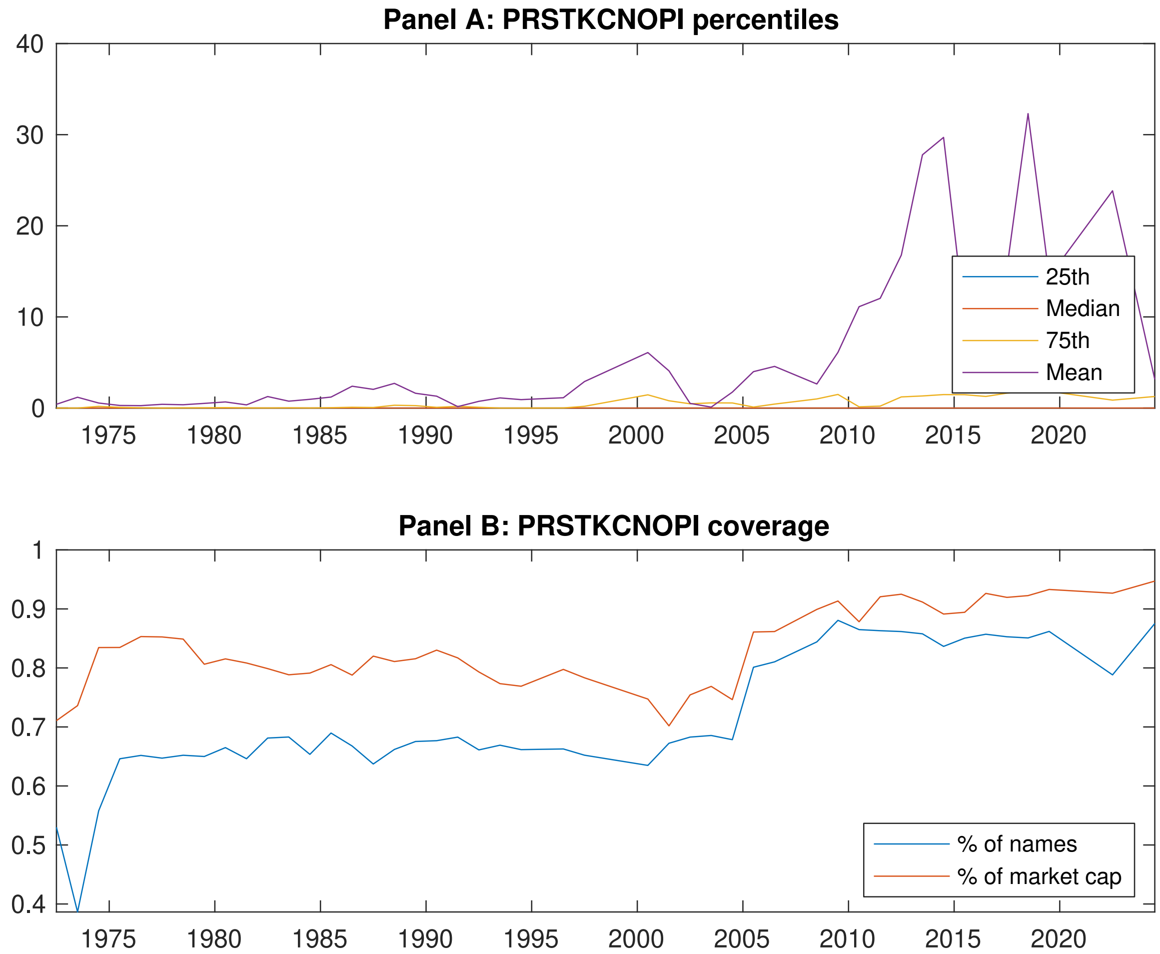


Figure 1: Times series of NBR percentiles and coverage.
This figure plots descriptive statistics for NBR. Panel A shows cross-sectional percentiles of NBR over the sample. Panel B plots the monthly coverage of NBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on NBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.63	0.52	0.60	0.84	0.37
	[2.34]	[2.97]	[2.70]	[3.43]	[4.65]	[4.19]
α_{CAPM}	-0.18	-0.05	-0.10	0.03	0.25	0.43
	[-2.98]	[-0.80]	[-1.73]	[0.61]	[4.96]	[4.94]
α_{FF3}	-0.25	0.01	-0.12	0.02	0.27	0.51
	[-4.45]	[0.25]	[-1.98]	[0.54]	[5.42]	[6.38]
α_{FF4}	-0.22	0.03	-0.13	0.03	0.26	0.48
	[-3.86]	[0.57]	[-2.11]	[0.63]	[5.15]	[5.81]
α_{FF5}	-0.21	0.12	-0.12	-0.01	0.15	0.36
	[-3.71]	[2.02]	[-2.07]	[-0.29]	[3.27]	[4.62]
α_{FF6}	-0.19	0.13	-0.13	-0.01	0.15	0.34
	[-3.29]	[2.18]	[-2.16]	[-0.13]	[3.28]	[4.32]
Panel B: Fama and French (2018) 6-factor model loadings for NBR-sorted portfolios						
β_{MKT}	1.04	1.02	0.99	0.95	0.99	-0.05
	[77.58]	[74.30]	[70.26]	[86.32]	[91.60]	[-2.73]
β_{SMB}	0.08	0.11	0.10	-0.09	-0.05	-0.13
	[4.06]	[5.31]	[4.53]	[-5.55]	[-2.85]	[-4.62]
β_{HML}	0.19	-0.15	0.01	-0.02	-0.12	-0.32
	[7.62]	[-5.66]	[0.53]	[-0.77]	[-5.92]	[-9.01]
β_{RMW}	-0.05	-0.24	0.01	0.06	0.23	0.28
	[-1.76]	[-9.03]	[0.26]	[3.02]	[11.10]	[7.79]
β_{CMA}	-0.08	-0.08	0.03	0.07	0.17	0.25
	[-2.01]	[-1.94]	[0.72]	[2.31]	[5.53]	[4.71]
β_{UMD}	-0.03	-0.02	0.01	-0.01	-0.00	0.03
	[-2.37]	[-1.11]	[0.74]	[-0.99]	[-0.31]	[1.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	744	899	739	534	490	
me ($\$10^6$)	1728	1376	1307	2866	3399	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the NBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37	0.43	0.51	0.48	0.36	0.34
			[4.19]	[4.94]	[6.38]	[5.81]	[4.62]	[4.32]
Quintile	NYSE	EW	0.24	0.29	0.28	0.26	0.12	0.11
			[2.95]	[3.56]	[3.74]	[3.39]	[1.68]	[1.55]
Quintile	Name	VW	0.28	0.34	0.43	0.39	0.30	0.27
			[3.17]	[3.87]	[5.41]	[4.82]	[3.80]	[3.46]
Quintile	Cap	VW	0.36	0.43	0.50	0.46	0.33	0.32
			[3.99]	[4.85]	[5.89]	[5.39]	[4.10]	[3.86]
Decile	NYSE	VW	0.29	0.32	0.42	0.36	0.36	0.31
			[2.79]	[3.08]	[4.23]	[3.58]	[3.50]	[3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.33	0.40	0.47	0.45	0.35	0.33
			[3.77]	[4.68]	[5.87]	[5.58]	[4.49]	[4.33]
Quintile	NYSE	EW	0.05	0.11	0.08	0.07		
			[0.57]	[1.26]	[1.02]	[0.92]		
Quintile	Name	VW	0.24	0.31	0.39	0.37	0.28	0.27
			[2.78]	[3.62]	[4.92]	[4.61]	[3.64]	[3.45]
Quintile	Cap	VW	0.33	0.41	0.46	0.44	0.33	0.32
			[3.61]	[4.64]	[5.50]	[5.25]	[4.11]	[3.98]
Decile	NYSE	VW	0.25	0.28	0.37	0.34	0.31	0.28
			[2.37]	[2.70]	[3.71]	[3.36]	[3.07]	[2.82]

Table 3: Conditional sort on size and NBR

This table presents results for conditional double sorts on size and NBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NBR and short stocks with low NBR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

	NBR Quintiles					NBR Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.66	0.62	0.59	0.56	0.86	0.20	0.21	0.20	0.23	0.15	0.18
	[2.42]	[2.20]	[2.12]	[2.08]	[3.48]	[2.06]	[2.10]	[2.00]	[2.30]	[1.52]	[1.79]
(2)	0.68	0.59	0.62	0.75	0.91	0.23	0.32	0.28	0.32	0.13	0.17
	[2.62]	[2.20]	[2.31]	[3.11]	[4.01]	[2.39]	[3.45]	[3.19]	[3.55]	[1.48]	[1.96]
(3)	0.68	0.56	0.66	0.82	0.91	0.23	0.29	0.25	0.29	0.11	0.15
	[2.87]	[2.18]	[2.74]	[3.65]	[4.26]	[2.44]	[3.11]	[2.85]	[3.18]	[1.26]	[1.71]
(4)	0.63	0.60	0.62	0.76	0.89	0.26	0.31	0.27	0.31	0.18	0.22
	[3.00]	[2.63]	[2.78]	[3.68]	[4.48]	[3.14]	[3.78]	[3.36]	[3.93]	[2.29]	[2.86]
(5)	0.47	0.53	0.51	0.72	0.76	0.29	0.35	0.45	0.41	0.29	0.28
	[2.36]	[2.66]	[2.97]	[3.95]	[4.20]	[2.83]	[3.44]	[4.56]	[4.12]	[3.04]	[2.82]

Panel B: Portfolio average number of firms and market capitalization

	NBR Quintiles Average n					NBR Quintiles Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	383	382	382	383	388	35	31	31	32	40
(2)	105	105	105	106	105	60	58	58	60	61
(3)	74	74	74	74	74	103	102	102	103	106
(4)	62	62	62	62	62	221	213	217	230	233
(5)	56	56	56	56	56	1388	1494	1860	2160	1676

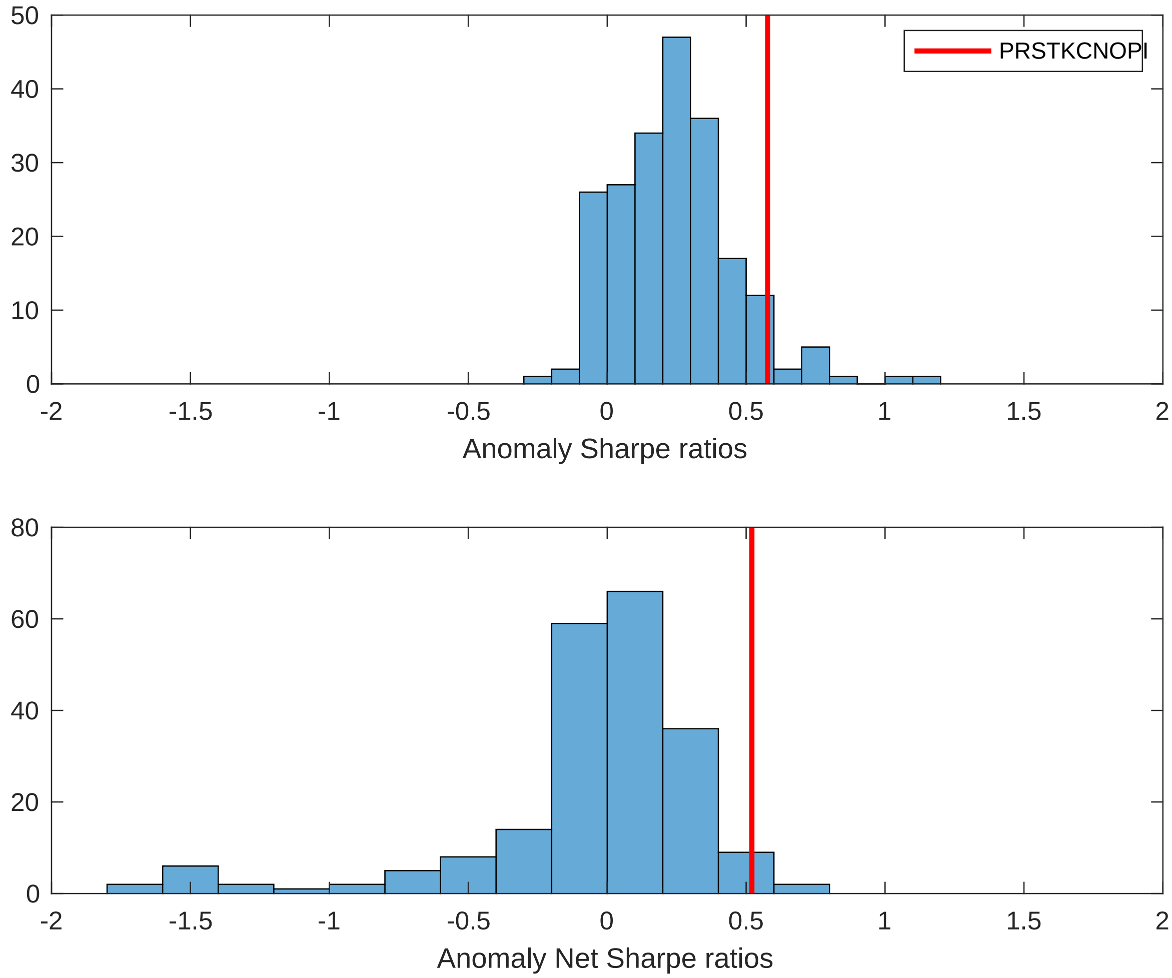


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

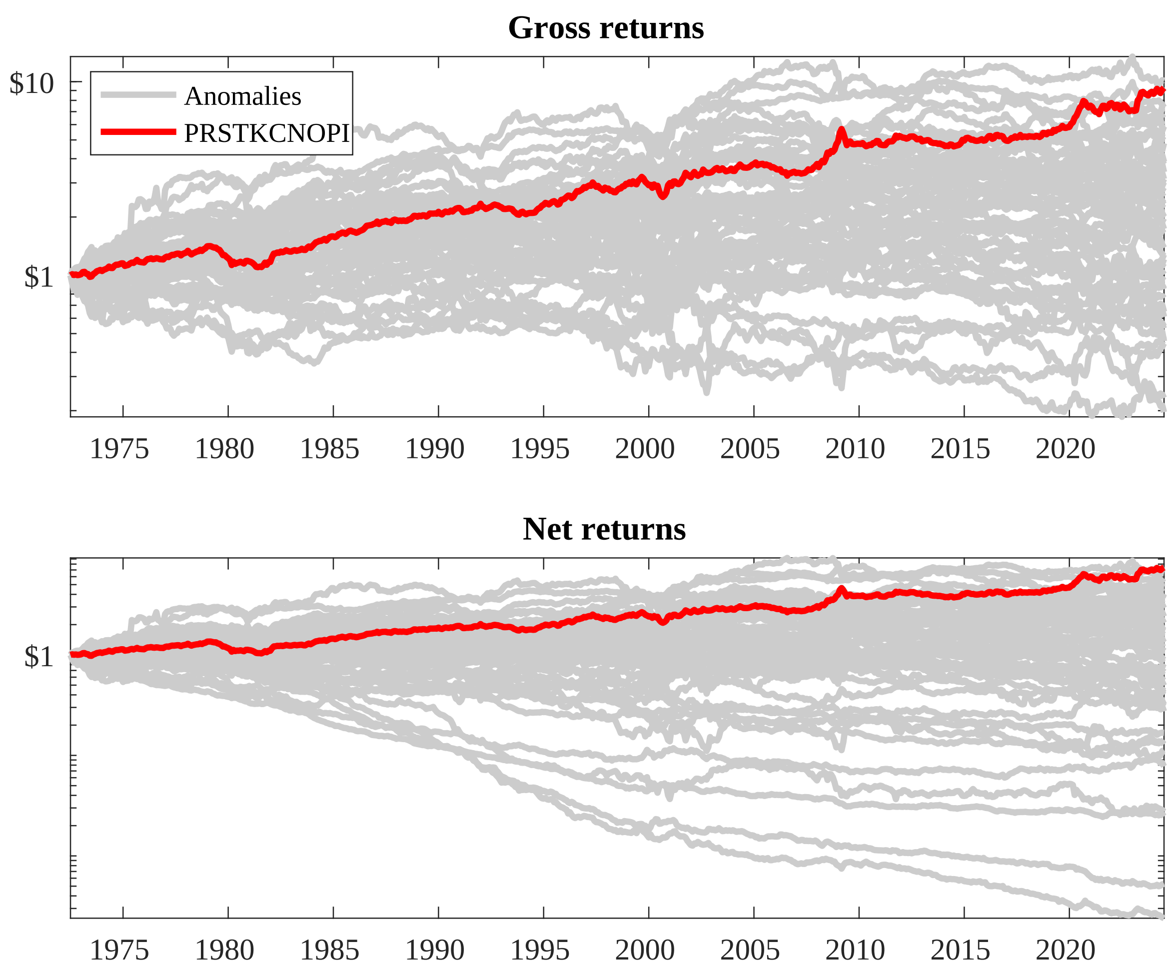
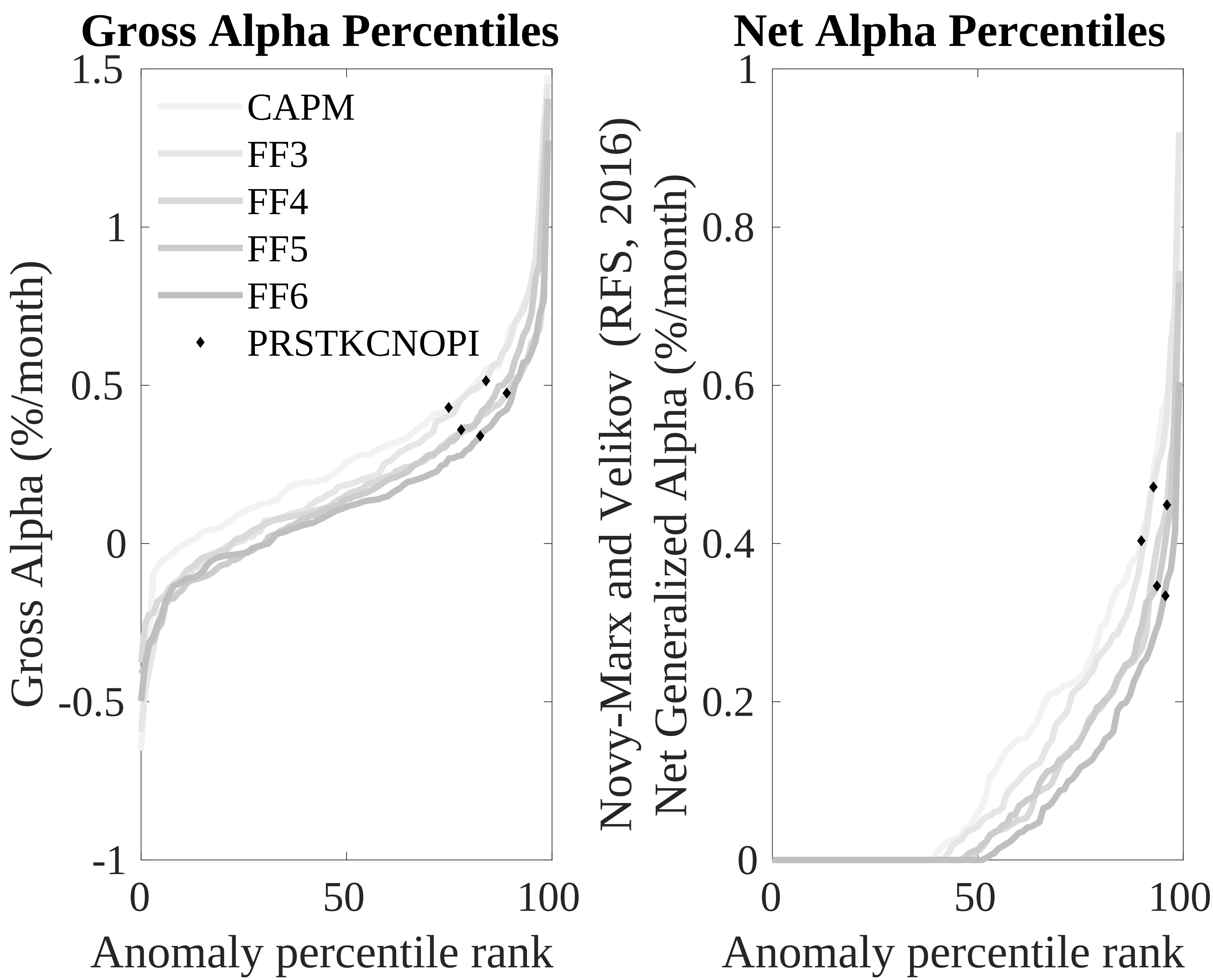


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
PRSTKCNOPI
1.5
1
0.5
0
-0.5
-1
0
50
100
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
1
0.8
0.6
0.4
0.2
0
0
50
100

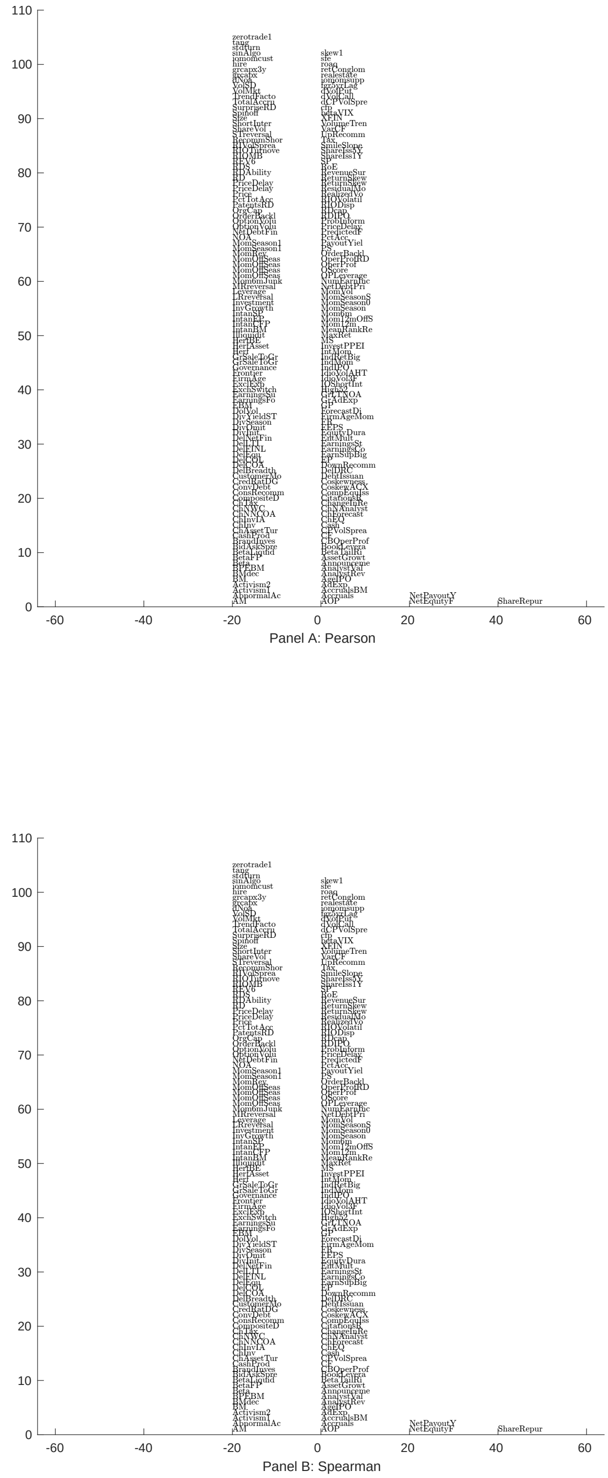


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with NBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

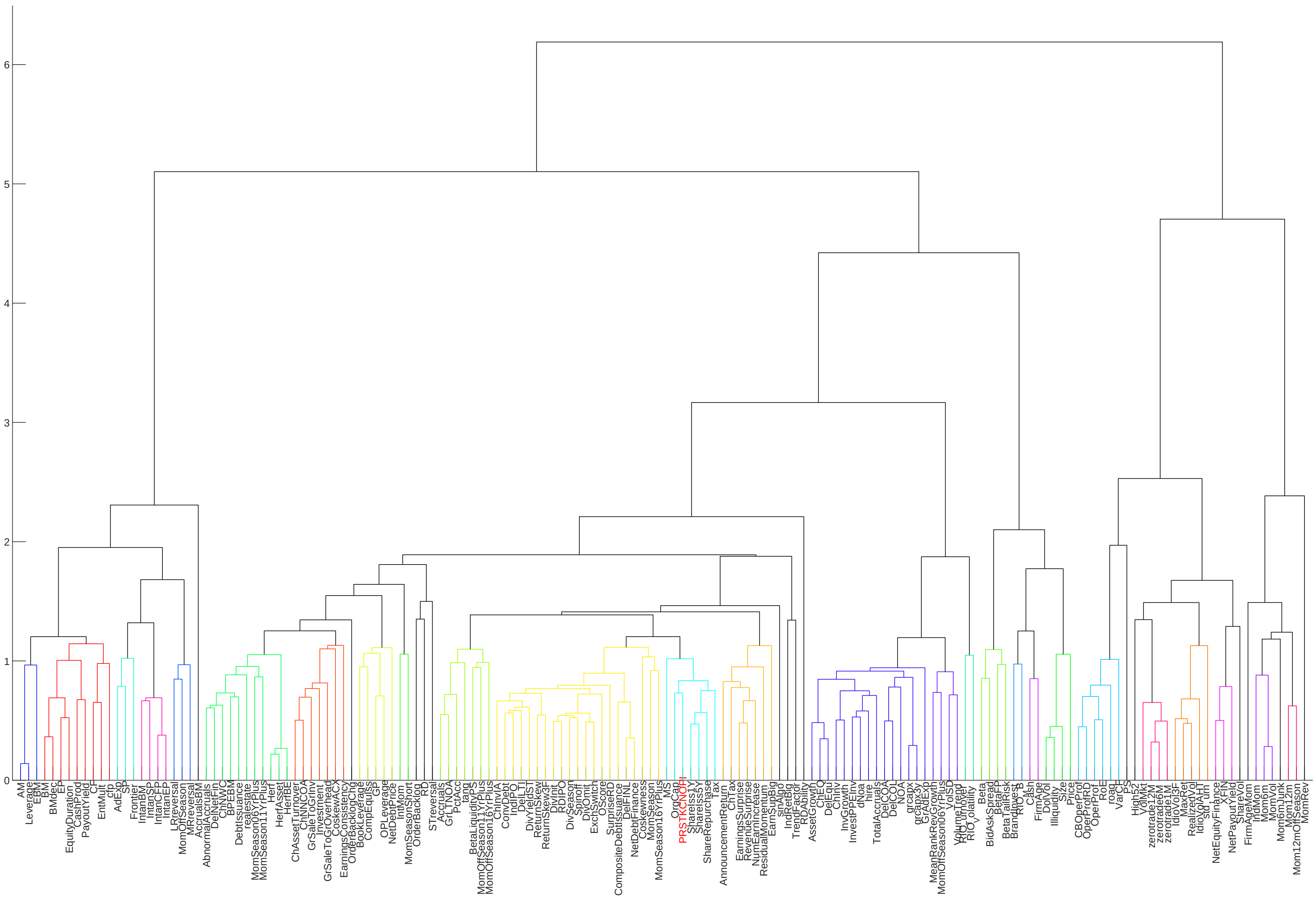


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

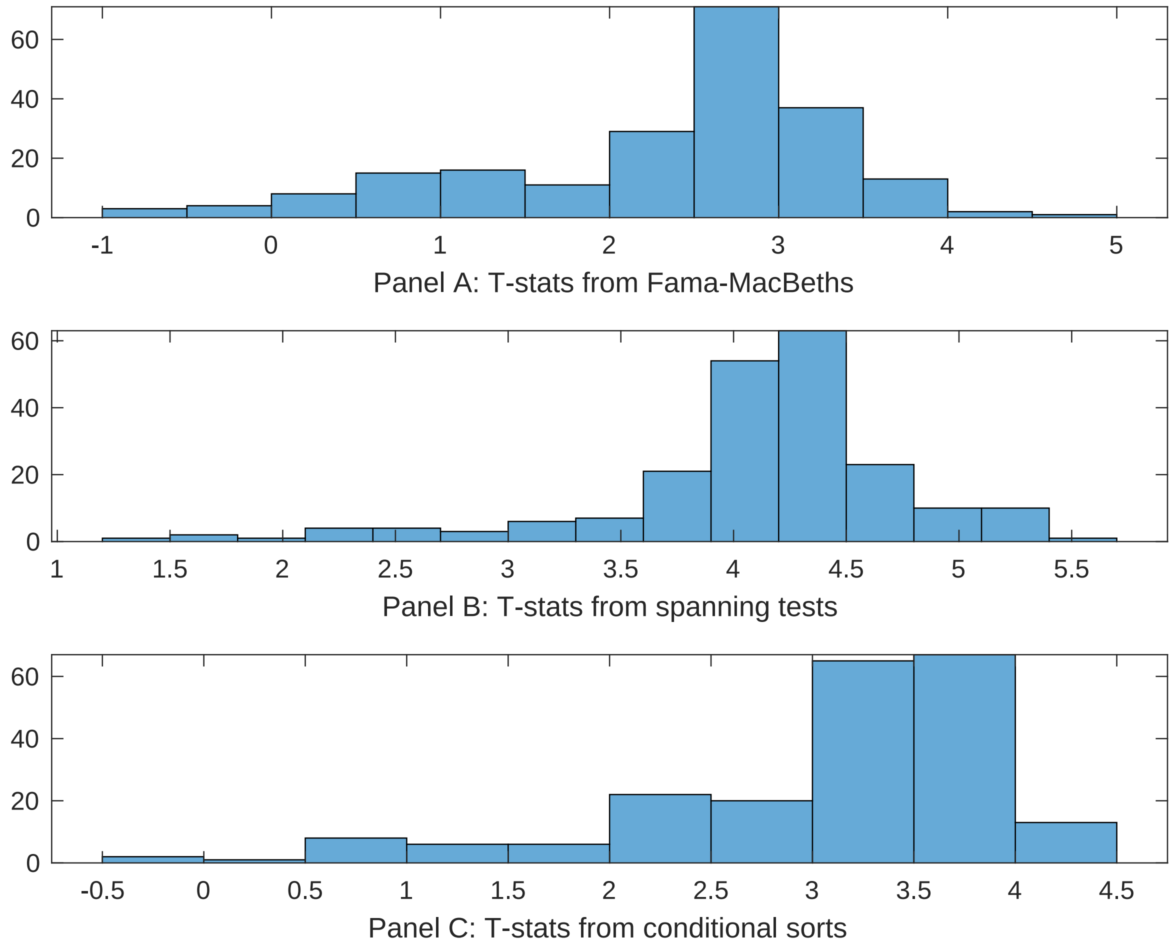


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR} NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NBR} NBR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12	0.11	0.11	0.10	0.11	0.99	0.11
	[5.28]	[3.74]	[4.62]	[3.93]	[4.57]	[3.45]	[3.88]
NBR	0.66	0.99	0.12	0.10	0.16	0.15	0.15
	[1.53]	[2.03]	[2.75]	[2.26]	[2.32]	[3.29]	[2.65]
Anomaly 1	0.19						0.90
	[7.16]						[3.44]
Anomaly 2		0.12					-0.35
		[2.96]					[-0.77]
Anomaly 3			-0.26				-0.11
			[-0.23]				[-0.86]
Anomaly 4				0.16			0.11
				[4.79]			[3.02]
Anomaly 5					0.65		0.40
					[3.07]		[2.74]
Anomaly 6						0.62	-0.37
						[1.14]	[-0.09]
# months	624	619	624	619	616	576	576
$\bar{R}^2$(%)	1	1	0	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the NBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NBR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.32	0.29	0.34	0.30	0.35	0.34	0.27
	[4.14]	[3.68]	[4.40]	[3.82]	[4.45]	[4.17]	[3.36]
Anomaly 1	24.50						23.34
	[6.38]						[5.25]
Anomaly 2		14.37					9.33
		[4.32]					[1.76]
Anomaly 3			13.55				7.04
			[3.07]				[1.36]
Anomaly 4				12.00			0.13
				[3.32]			[0.02]
Anomaly 5					2.78		-9.16
					[0.86]		[-2.32]
Anomaly 6						6.81	0.68
						[2.11]	[0.19]
mkt	-1.32	-2.33	-2.77	-3.75	-4.65	-3.76	0.16
	[-0.71]	[-1.23]	[-1.43]	[-2.03]	[-2.50]	[-1.85]	[0.08]
smb	-4.08	-7.18	-7.40	-8.24	-11.64	-12.10	-4.81
	[-1.40]	[-2.45]	[-2.39]	[-2.79]	[-4.16]	[-3.34]	[-1.29]
hml	-27.84	-25.72	-29.04	-26.89	-29.36	-34.64	-29.89
	[-8.15]	[-7.14]	[-8.28]	[-7.44]	[-7.97]	[-9.28]	[-7.33]
rmw	12.80	16.93	15.32	21.69	24.71	20.99	9.61
	[3.08]	[3.99]	[2.87]	[5.56]	[5.32]	[4.25]	[1.61]
cma	8.70	25.27	27.13	23.24	24.17	22.99	10.47
	[1.53]	[4.83]	[5.06]	[4.40]	[4.54]	[4.20]	[1.70]
umd	2.86	1.59	2.93	1.99	2.48	2.39	4.58
	[1.63]	[0.88]	[1.63]	[1.10]	[1.28]	[1.19]	[2.23]
# months	624	620	624	620	620	577	577
$\bar{R}^2(\%)$	32	30	28	29	27	32	38

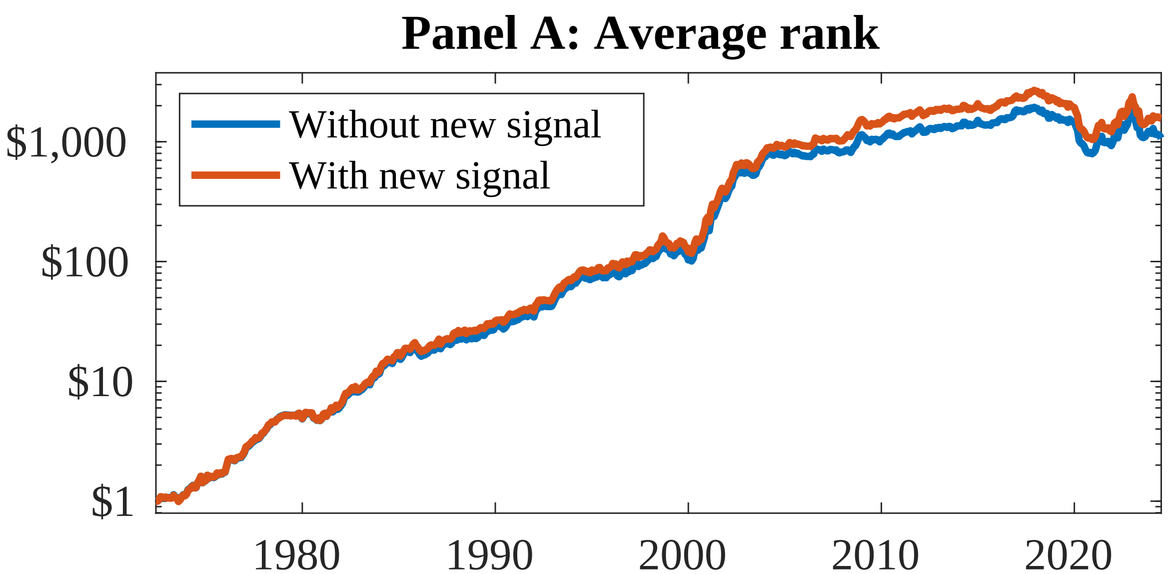


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as NBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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